An Analysis of Bias variance tradeoff

Bias variance tradeoff

$$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma_\varepsilon^2$$

Bias variance tradeoff -- Part 1

Derivation The derivation of the bias–variance decomposition for squared error proceeds as follows. For convenience, we drop the D {\displaystyle D} subscript in the following lines, such that f ^ (x ; D) = f ^ (x) {\displaystyle {\hat {f}}\!(x;D)={\hat {f}}\!(x)} . Let us write the mean-squared error of our model:

Bias variance tradeoff -- Part 2

In Monte Carlo methods While in traditional Monte Carlo methods the bias is typically zero, modern approaches, such as Markov chain Monte Carlo are only asymptotically unbiased, at best. Convergence diagnostics can be used to control bias via burn-in removal, but due to a limited computational budget, a bias–variance trade-off arises, leading to a wide range of approaches, in which a controlled bias is accepted, if this allows to dramatically reduce the variance, and hence the overall estimation error.

Bias variance tradeoff -- Part 3

In classification The bias–variance decomposition was originally formulated for least-squares regression. For the case of classification under the 0-1 loss (misclassification rate), it is possible to find a similar decomposition, with the caveat that the variance term becomes dependent on the target label. Alternatively, if the classification problem can be phrased as probabilistic classification, then the expected cross-entropy can instead be decomposed to give bias and variance terms with the same semantics but taking a different form. It has been argued that as training data increases, the variance of learned models will tend to decrease, and hence that as training data quantity increases, error is minimised by methods that learn models with lesser bias, and that conversely, for smaller training data quantities it is ever more important to minimise variance.

Bias variance tradeoff -- Part 4

σ 2 = E y ■ [(y − f (x) ■ E y | x [y]) 2] {\displaystyle \sigma ^{2}=\operatorname {E} _{y}{\Big [}{\big (}y-\underbrace {f(x)} _{E_{y|x}[y]}{\big)}^{2}{\Big]}}

Bias variance tradeoff -- Part 5

the square of the bias of the learning method, which can be thought of as the error caused by the simplifying assumptions built into the method. E.g., when approximating a non-linear function f (x) {\displaystyle f(x)} using a learning method for linear models, there will be error in the estimates f ^ (x) {\displaystyle {\hat {f}}\!(x)} due to this assumption; the variance of the learning method, or, intuitively, how much the learning method f ^ (x) {\displaystyle {\hat {f}}\!(x)} will move around its mean; the irreducible error σ 2 {\displaystyle \sigma ^{2}} . Since all three terms are non-negative, the irreducible error forms a lower bound on the expected error on unseen samples. The more complex the model f ^ (x) {\displaystyle {\hat {f}}\!(x)} is, the more data points it will capture, and the lower the bias will be. However, complexity will make the model "move" more to capture the data points, and hence its variance will be larger.

Bias variance tradeoff -- Part 6

linear and Generalized linear models can be regularized to decrease their variance at the cost of increasing their bias. In artificial neural networks, the variance increases and the bias decreases as the number of hidden units increase, although this classical assumption has been the subject of recent debate. Like in GLMs, regularization is typically applied. In k-nearest neighbor models, a high value of k leads to high bias and low variance (see below). In instance-based learning, regularization can be achieved varying the mixture of prototypes and exemplars. In decision trees, the depth of the tree determines the variance. Decision trees are commonly pruned to control variance. One way of resolving the trade-off is to use mixture models and ensemble learning. For example, boosting combines many "weak" (high bias) models in an ensemble that has lower bias than the individual models, while bagging combines "strong" learners in a way that reduces their variance. Model validation methods such as cross-validation (statistics) can be used to tune models so as to optimize the trade-off.

Bias variance tradeoff -- Part 7

Var D ■ [f ^ (x ; D)] ■ E D [(E D [f ^ (x ; D)] − f ^ (x ; D)) 2] {\displaystyle \operatorname {Var} _{D}{\big [}{\hat {f}}\!(x;D){\big]}\triangleq \mathbb {E} _{D}\left[\left(\mathbb {E} _{D}[{\hat {f}}\!(x;D)]-{\hat {f}}\!(x;D)\right)^{2}\right]}

Bias variance tradeoff -- Part 8

In regression The bias–variance decomposition forms the conceptual basis for regression regularization methods such as LASSO and ridge regression. Regularization methods introduce bias into the regression solution that can reduce variance considerably relative to the ordinary least squares (OLS) solution. Although the OLS solution provides non-biased regression estimates, the lower variance solutions produced by regularization techniques provide superior MSE performance.

Bias variance tradeoff -- Part 9

The bias–variance tradeoff is a central problem in supervised learning. Ideally, one wants to choose a model that both accurately captures the regularities in its training data, but also generalizes well to unseen data. Unfortunately, it is typically impossible to do both simultaneously. High-variance learning methods may be able to represent their training set well but are at risk of overfitting to noisy or unrepresentative training data. In contrast, algorithms with high bias typically produce simpler models that may fail to capture important regularities (i.e. underfit) in the data. It is an often made fallacy to assume that complex models must have high variance. High variance models are "complex" in some sense, but the reverse need not be true. In addition, one has to be careful how to define complexity. In particular, the number of parameters used to describe the model is a poor measure of complexity. The model f a , b (x) = a sin ■ (b x) {\displaystyle f_{a,b}(x)=a\sin(bx)} has only two parameters (a , b {\displaystyle a,b}) but it can interpolate any number of points by oscillating with a high enough frequency, resulting in both a high bias and high variance. An analogy can be made to the relationship between accuracy and precision. Accuracy is one way of quantifying bias and can intuitively be improved by selecting from only local information. Consequently, a sample will appear accurate (i.e. have low bias) under the aforementioned selection conditions, but may result in underfitting. In other words, test data may not agree as closely with training data, which would indicate imprecision and therefore inflated variance. A graphical example would be a straight line fit to data exhibiting quadratic behavior overall. Precision is a description of variance and generally can only be improved by selecting information from a comparatively larger space. The option to select many data points over a broad sample space is the ideal condition for any analysis. However, intrinsic constraints (whether physical, theoretical, computational, etc.) will always play a limiting role. The limiting case where only a finite number of data points are selected over a broad sample space may result in improved precision and lower variance overall, but may also result in an overreliance on the training data (overfitting). This means that test data would also not agree as closely with the training data, but in this case the reason is inaccuracy or high bias. To borrow from the previous example, the graphical representation would appear as a high-order polynomial fit to the same data exhibiting quadratic behavior. Note that error in each case is measured the same way, but the reason ascribed to the error is different depending on the balance between bias and variance. To mitigate how much information is used from neighboring observations, a model can be smoothed via explicit regularization, such as shrinkage.